Artificial Intelligence

Problem Solving by Informed Search

Conducted by:

Dr. Fazlul Hasan Siddiqui

(fazlul.siddiqui@duet.ac.bd)

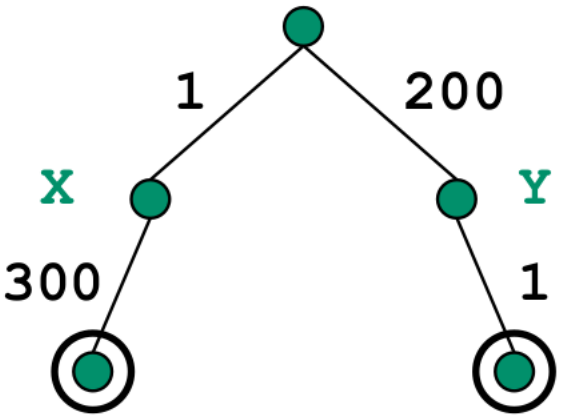
Credit: MIT, USA

Uninformed vs. Informed Search

- Depth-first, breadth-first and uniform-cost searches are uninformed.
- In informed search there is an estimate available of the cost (distance) from each state (city) to the goal.
- This estimate (heuristic) can help you head in the right direction.
- Heuristic embodied in function h(n), estimate of remaining cost from search node n to the least cost goal.
- Graph being searched is a graph of states. Search algorithm defines a tree of search nodes. Two paths to the same state generate two different search nodes.
- Heuristic could be defined on underlying state; the path to a state does not affect estimate of distance to the goal.

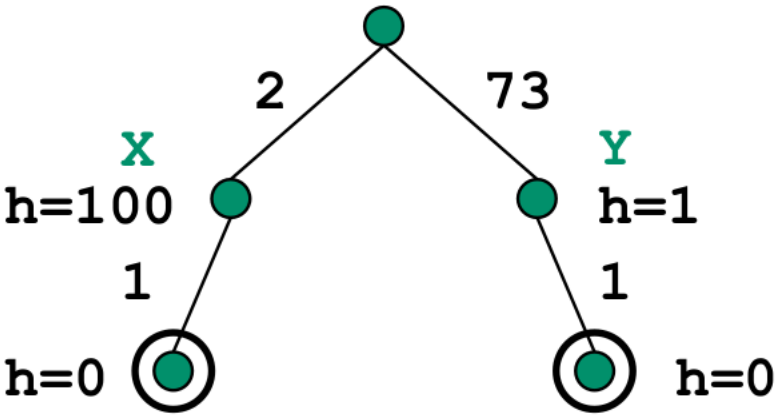
Using Heuristic Information

- Should we go to X or Y?
- Uniform cost says go to X
- If $h(X) \gg h(Y)$, this should affect our choice
- If g(n) is path-length of node n, we can use g(n) + h(n) to prioritize the agenda
- This method is called A^* [pronounced "A star"]



Admissibility

- What must be true about h for A^* to find optimal path?
- A^* finds optimal path if h is admissible; h is admissible when it never overestimates.
- In this example, h is not admissible.
- In route finding problems, straight-line distance to goal is admissible heuristic.

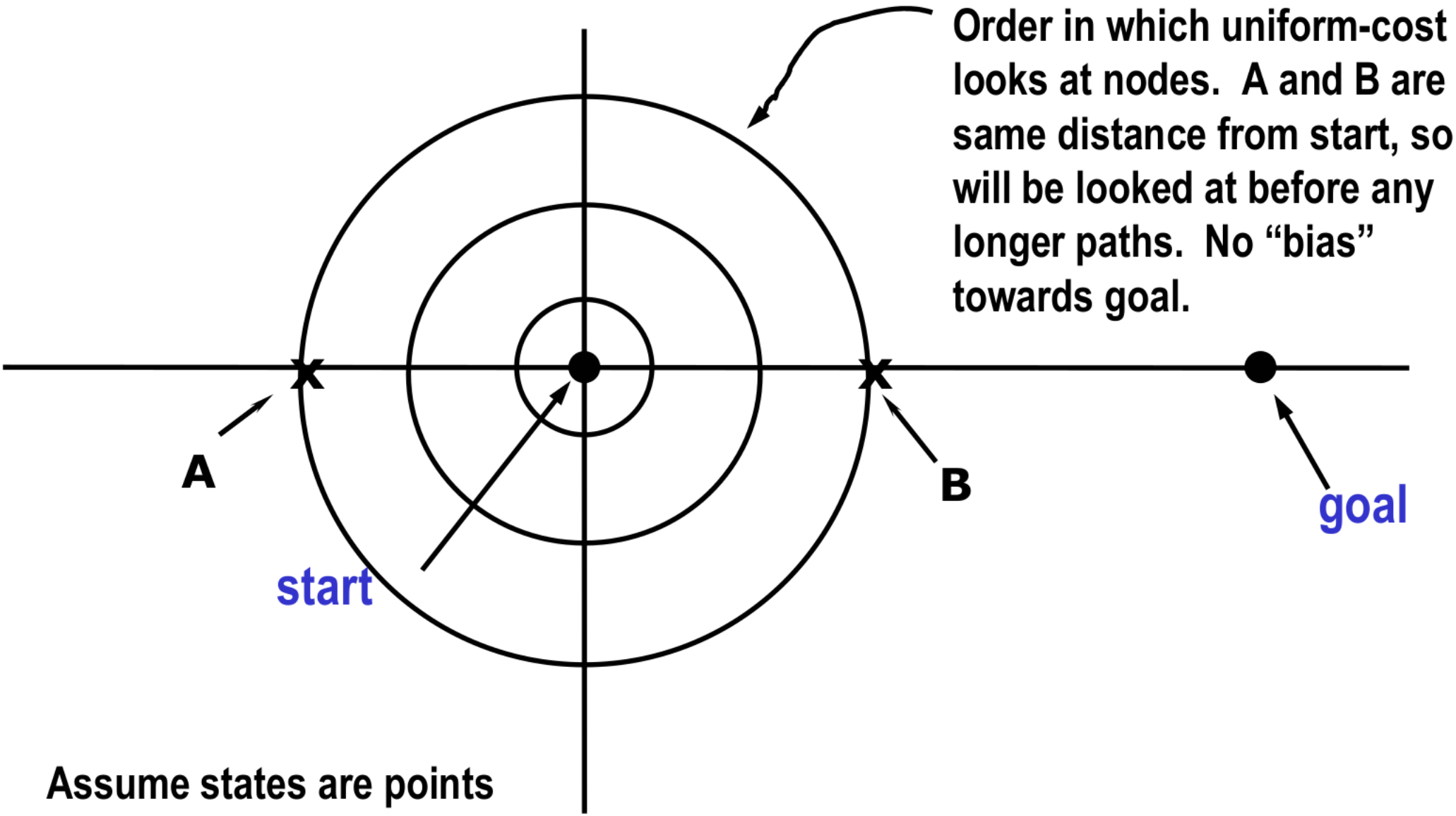


g(X)+h(X) = 102

g(Y)+h(Y) = 74

Optimal path is not found!

Why use estimate of goal distance?



Assume states are points the Euclidean plane.

A* search: Minimizing the total estimated solution cost

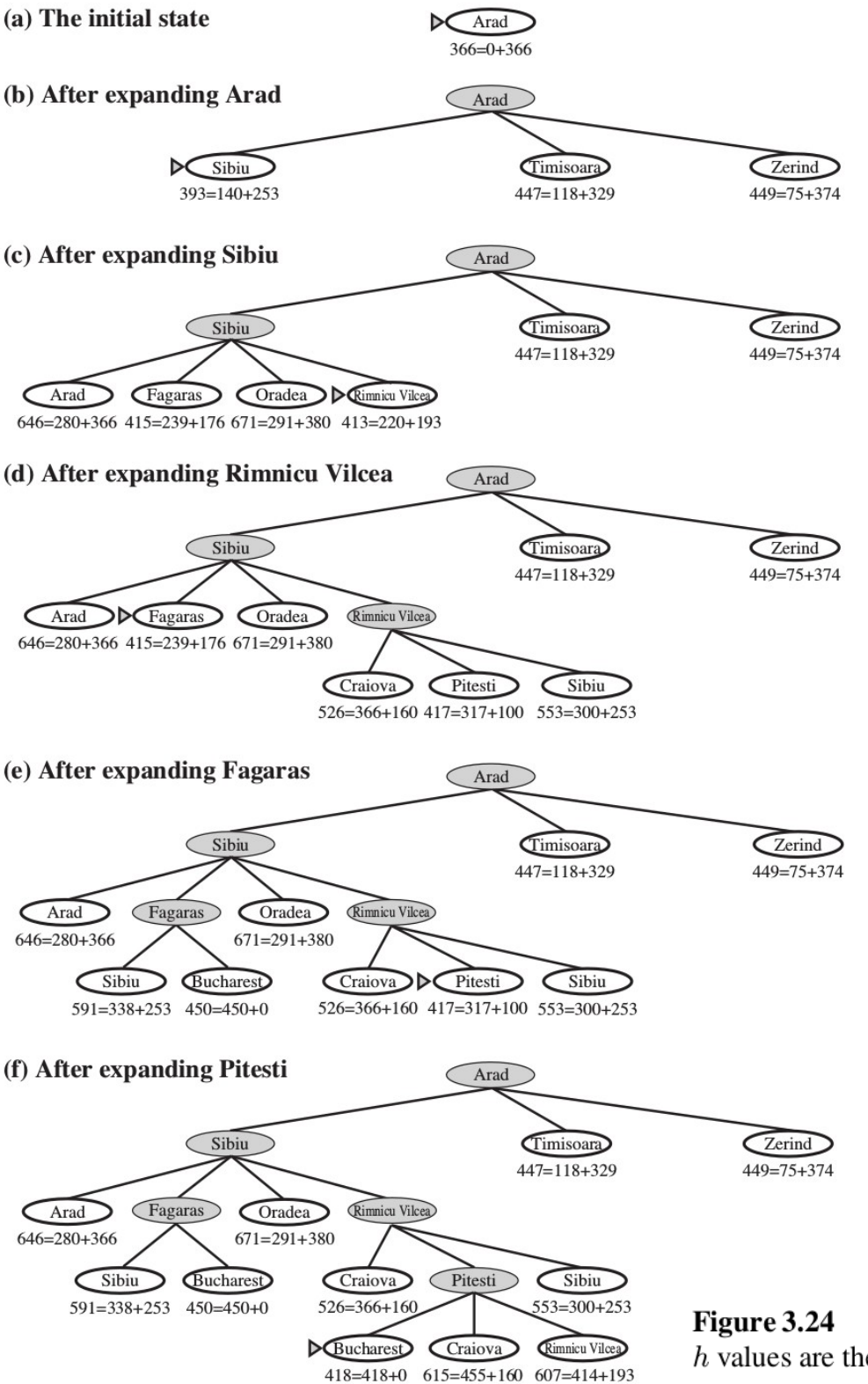
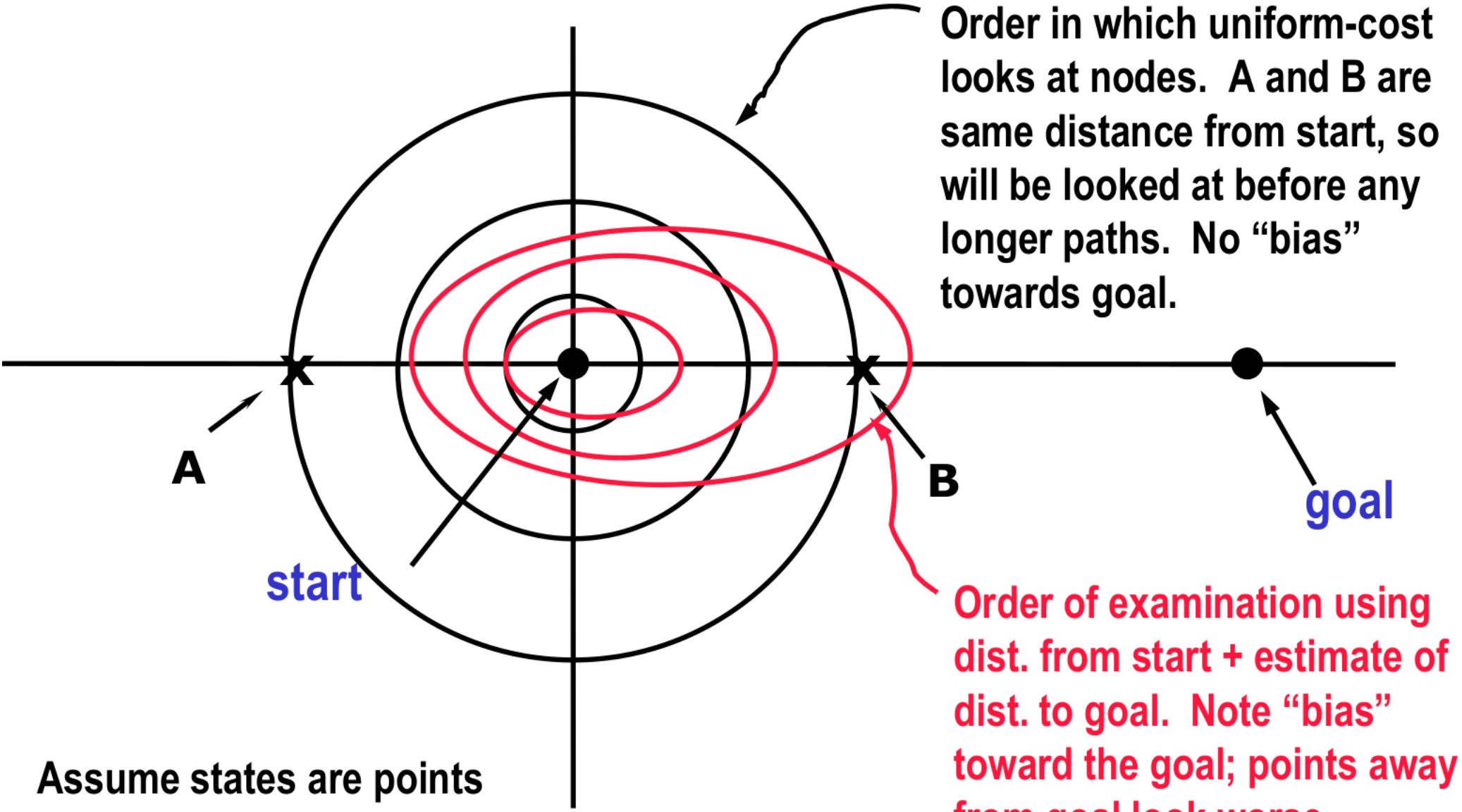


Figure 3.24 Stages in an A* search for Bucharest. Nodes are labeled with $f = g + h$. The h values are the straight-line distances to Bucharest taken from Figure 3.22.

Why use estimate of goal distance?



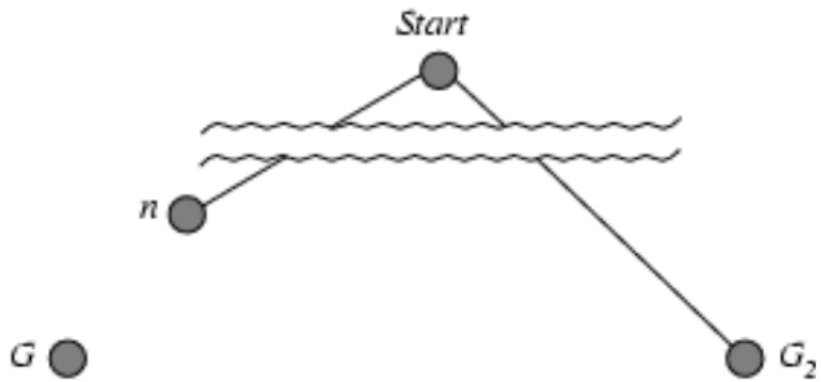
Assume states are points the Euclidean plane.

Admissible heuristics

- A heuristic $h(n)$ is admissible if for every node n, $h(n) \leq h^*(n)$, where $h^*(n)$ is the true cost to reach the goal state from n.
- An admissible heuristic never overestimates the cost to reach the goal, i.e., it is optimistic
- Example: $h_{SLD}(n)$ (never overestimates the actual road distance)
- Theorem: If $h(n)$ is admissible, A* using TREE-SEARCH is optimal

Optimality of A* (proof)

- Suppose some suboptimal goal G_2 has been generated and is in the fringe. Let *n* be an unexpanded node in the fringe such that *n* is on a shortest path to an optimal goal *G*.



- $f(G_2) = g(G_2)$ since $h(G_2) = 0$
- $g(G_2) > g(G)$ since G_2 is suboptimal
- $f(G) = g(G)$ since $h(G) = 0$
- $f(G_2) > f(G)$ from above
- $h(n) \leq h^*(n)$ since h is admissible
- $g(n) + h(n) \leq g(n) + h^*(n)$
- $f(n) \leq f(G)$
- Hence f(G2) > f(n), and A* will never select G2 for expansion

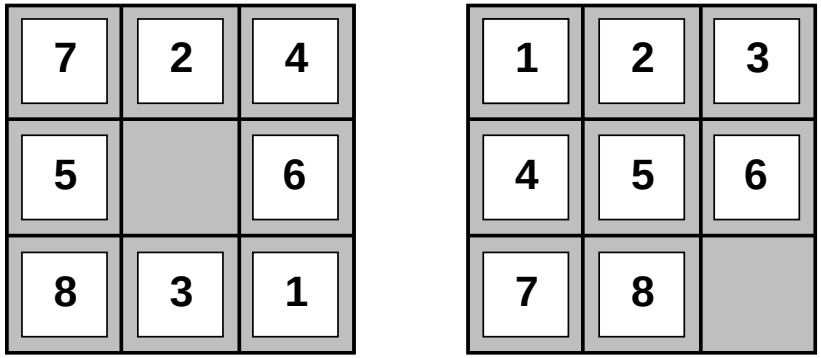
Heuristics

- If we set h=0, then A^* is uniform-cost search; h=0 is admissible heuristic (when all costs are non-negative).
- Very difficult to find heuristics that guarantee sub-exponential worst-case cost.
- Heuristic functions can be solutions to "relaxed" version of original problem, e.g. straight line distance is solution to route-finding when we relax constraint to follow the roads.

Admissible heuristics

E.g., for the 8-puzzle:

$h_1(n)$ = number of misplaced tiles
$h_2(n)$ = total Manhattan distance
(i.e., no. of squares from desired location of each tile)



Start State **Goal State**

$h_1(S) =$?? 6
$h_2(S) =$?? 4+0+3+3+1+0+2+1 = 14

Dominance

Given two admissible heuristics h_a, h_b,
if $h_b(n) \geq h_a(n)$ for all n then h_b dominates h_a and is better for search

In the 8-puzzle h_2 dominates h_1. Typical search costs:

$d = 14$ IDS = 3,473,941 nodes
A$^*(h_1)$ = 539 nodes
A$^*(h_2)$ = 113 nodes
$d = 24$ IDS $\approx$ 54,000,000,000 nodes
A$^*(h_1)$ = 39,135 nodes
A$^*(h_2)$ = 1,641 nodes

Given two admissible heuristics h_a, h_b,

$$h(n) = \max(h_a(n), h_b(n))$$

is also admissible and dominates h_a, h_b

Relaxed problems

Admissible heuristics can be derived from the **optimal** solution cost of a **relaxed** version of the problem

Rules of the 8-puzzle:

a tile can move from square A to square B if
A is adjacent to B **and** B is blank;

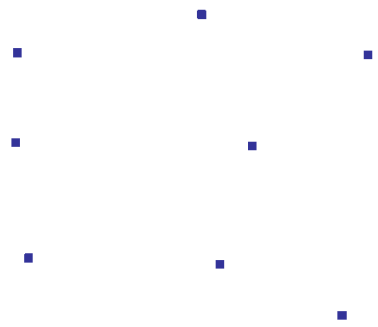
Relaxations:

- ♢ a tile can move from square A to square B if A is adjacent to B
- ♢ a tile can move from square A to square B if B is blank
- ♢ a tile can move from square A to square B

Key point: the optimal solution cost of a relaxed problem is no greater than the optimal solution cost of the real problem

Relaxed problems contd.

Well-known example: travelling salesperson problem (TSP)
Find the shortest tour visiting all cities exactly once and returning to first city



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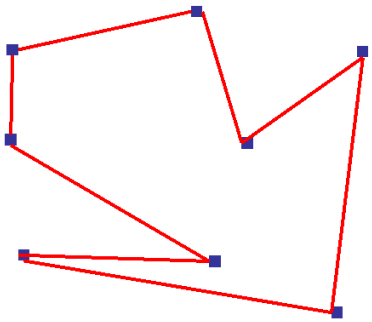

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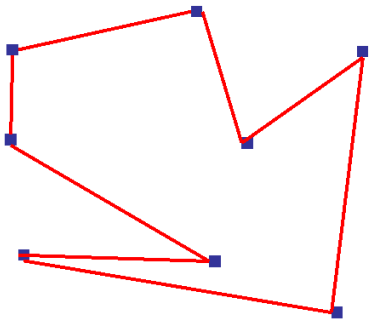

The set of arcs which covers all nodes at minimum cost
and forms a tour

Relaxed problems contd.

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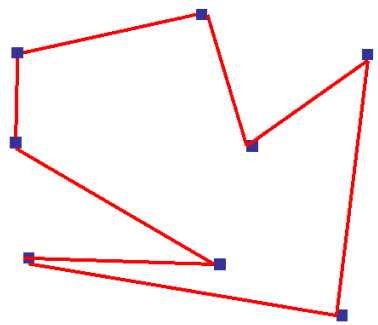
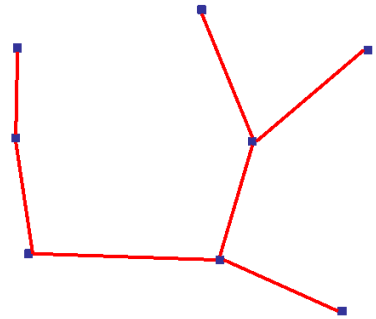


The set of arcs which covers all nodes at minimum cost
so that each node has degree 2

Relaxed problems contd.

Minimum spanning tree (MST)
The set of arcs which covers all nodes at minimum cost



Minimum spanning tree can be computed in $O(n^2)$
and is a lower bound on the shortest (open) tour

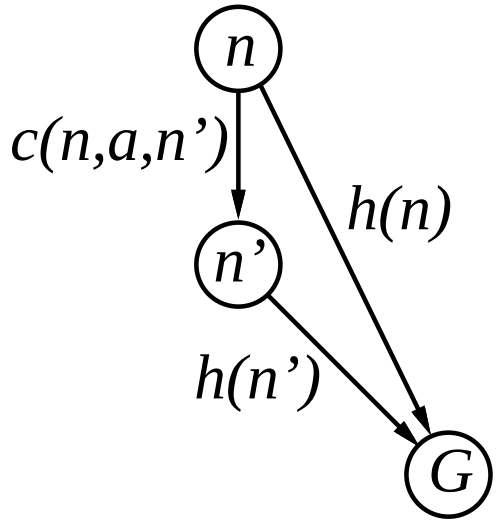
Consistency

A heuristic is consistent if

$$h(n) \leq c(n, a, n') + h(n')$$

If h is consistent, then h is admissible, and $f(n)$ is nondecreasing along any path:

$$\begin{aligned} f(n') &= g(n') + h(n') \\ &= g(n) + c(n, a, n') + h(n') \\ &\geq g(n) + h(n) \\ &= f(n) \end{aligned}$$

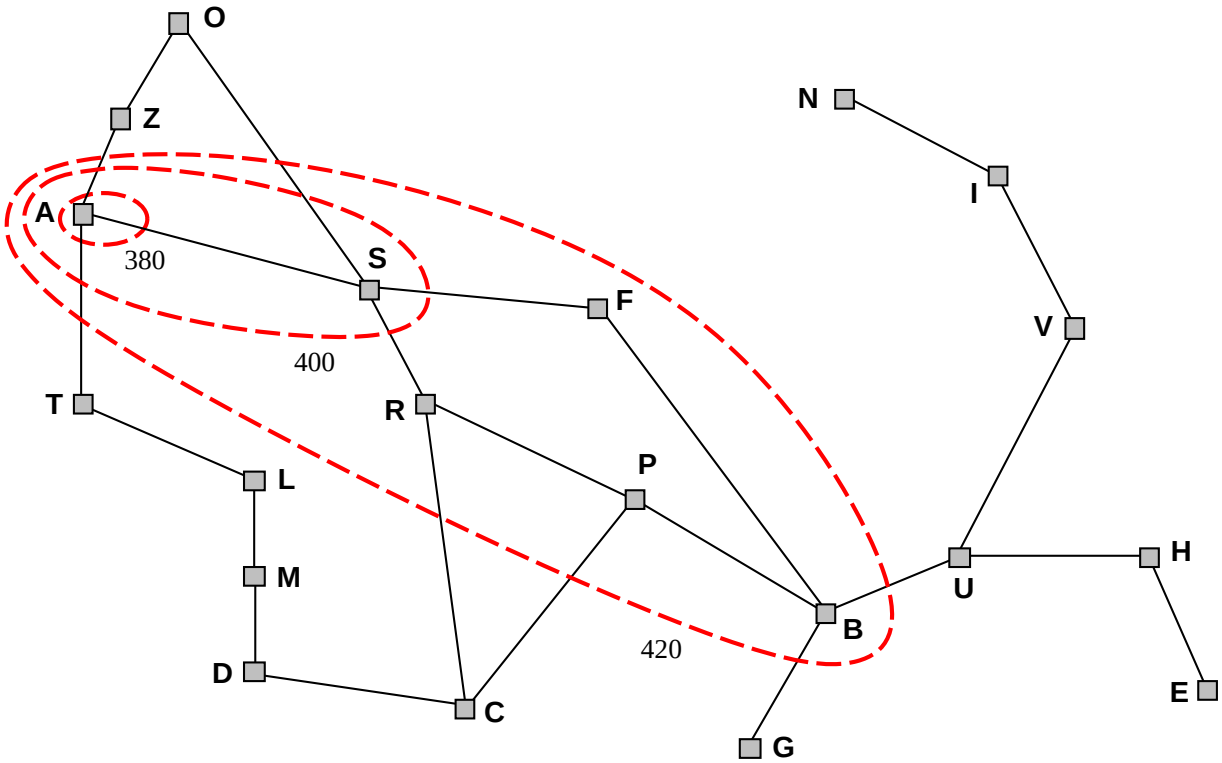


Consequently, when expanding a node, we cannot get a node with a smaller f, and so the value of the best node on the fringe will never decrease.

Optimality of A* (based on consistency)

Consistency: A* expands nodes in order of increasing f value

Gradually expands "f-contours" of nodes (cf. breadth-first expands layers, uniform-cost expands g-contours)
Contour i has all nodes with $f = f_i$, where $f_i < f_{i+1}$



Properties of A*

Complete?? Yes, unless there are infinitely many nodes with $f \leq f(G)$

Time?? Exponential in [relative error in h $\times$ length of soln.]

Space?? Keeps all nodes in memory

Optimal?? Yes—cannot expand f_{i+1} until f_i is finished

A* expands all nodes with $f(n) < C^*$
A* expands some nodes with $f(n) = C^*$
A* expands no nodes with $f(n) > C^*$

(Heuristic) Iterative Deepening: IDA*

- Like Iterative Deepening DFS
  - But the depth bound is measured in terms of the f value
- If you don't find a solution at a given depth
  - Increase the depth bound:
    to the minimum of the f-values that exceeded the previous bound

Best-first search

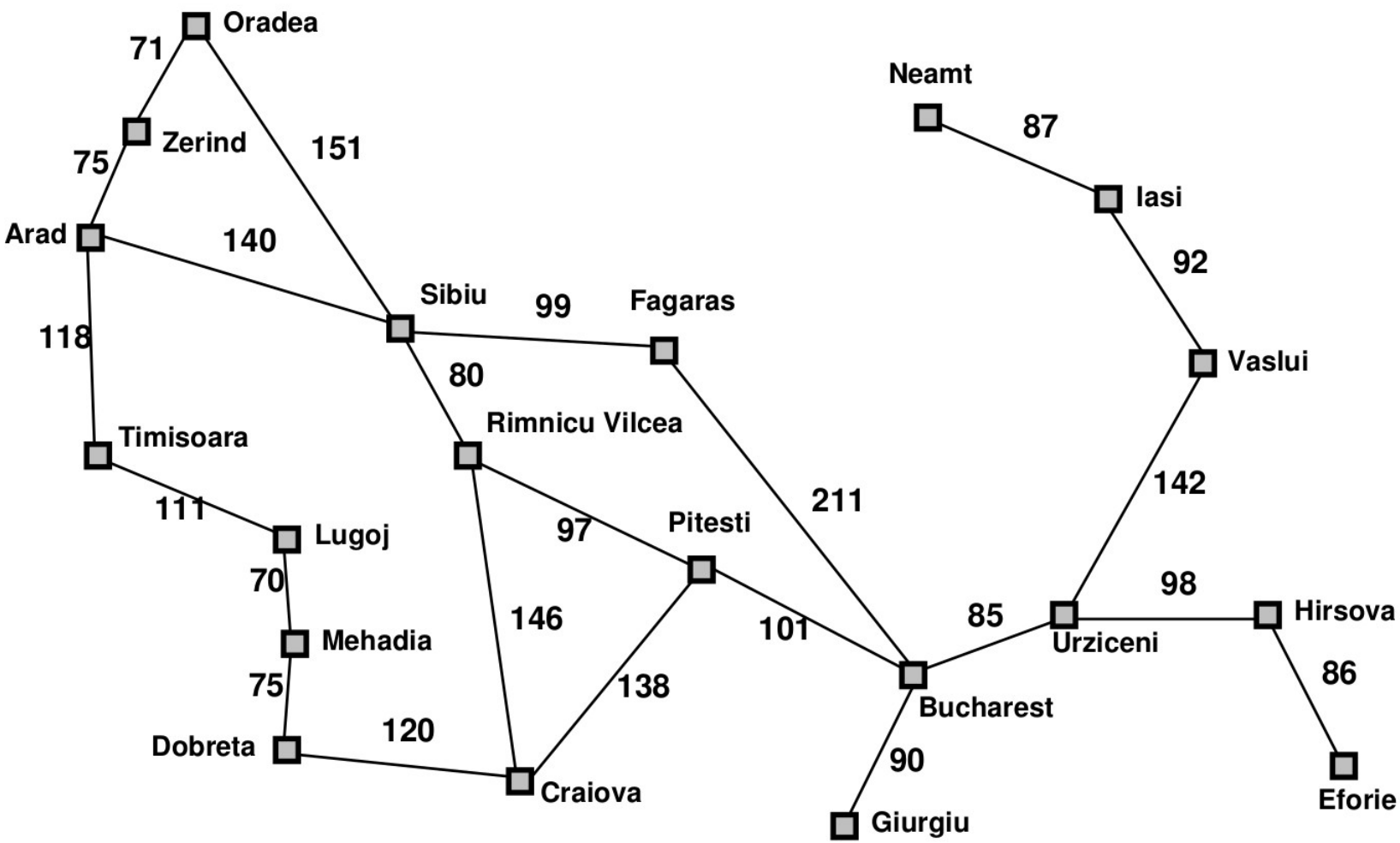
- The general approach we will consider is called **best-first search.** Best-first search is an instance of the general TREE-SEARCH or GRAPH-SEARCH algorithm in which a node is selected for expansion based on an **evaluation function,** f (n) .

- <u>Implementation</u>:
  - Best-first search can be implemented within our general search framework via a priority queue, a data structure that will maintain the fringe in ascending order of f –values.
  - A key component of these algorithms is a heuristic function, denoted by h(n):

    h(n) = estimated cost of the cheapest path from node n to a goal node.

    if *n* is a goal node, then *h(n)* = 0.

Romania with step costs in km



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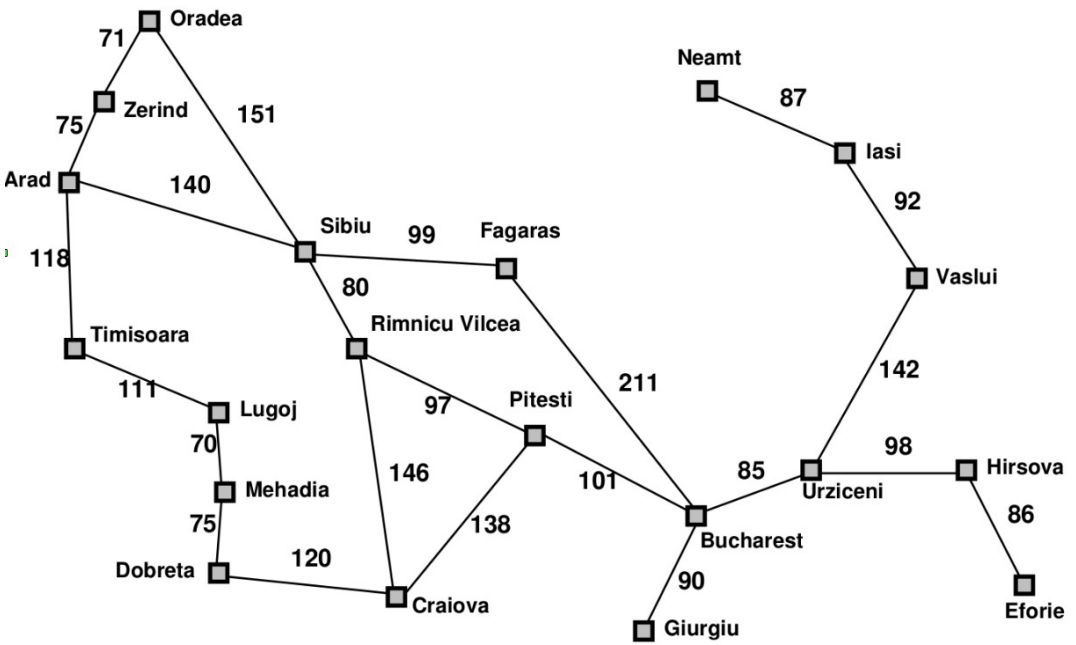
Greedy best-first search

- Greedy best-first search tries to expand the node that is closest to the goal.
- Evaluation function $f(n) = h(n)$ (heuristic)
- e.g., $h_{SLD}(n)$ = straight-line distance from n to Bucharest
- Greedy best-first search expands the node that appears to be closest to goal

Greedy Best-first search



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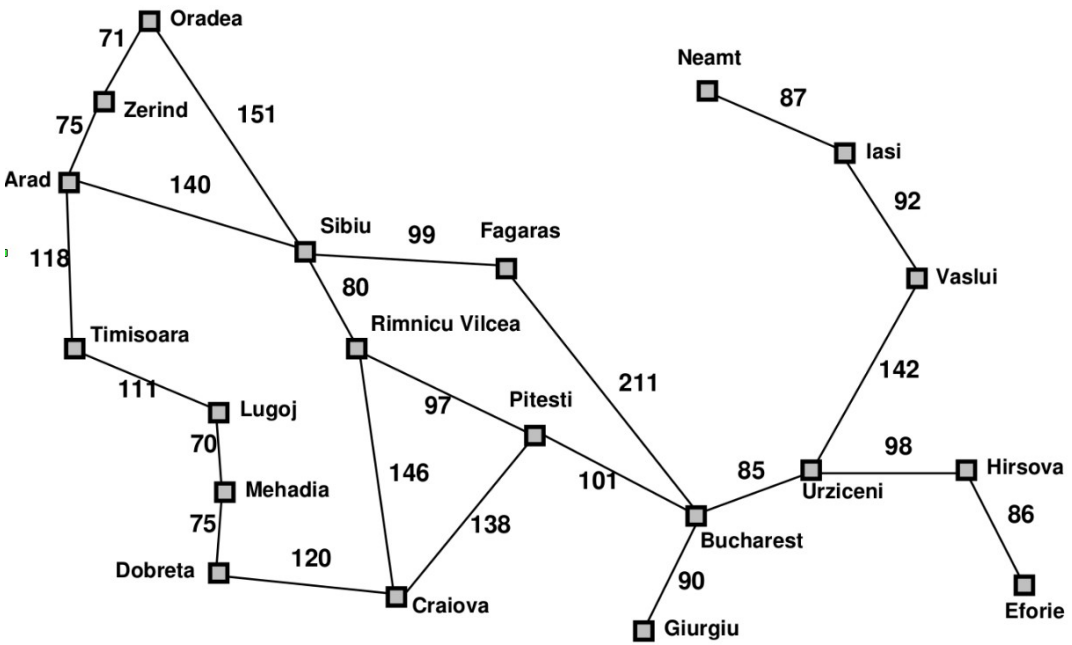
Greedy Best-first search



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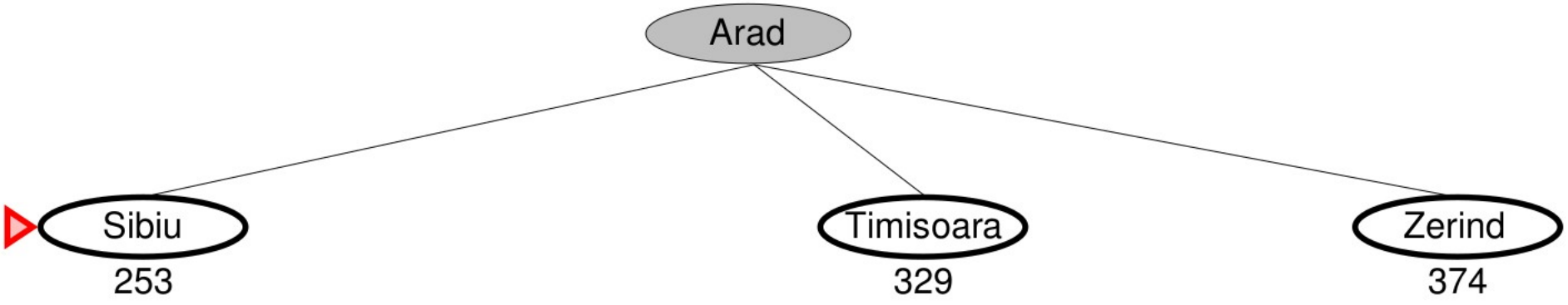

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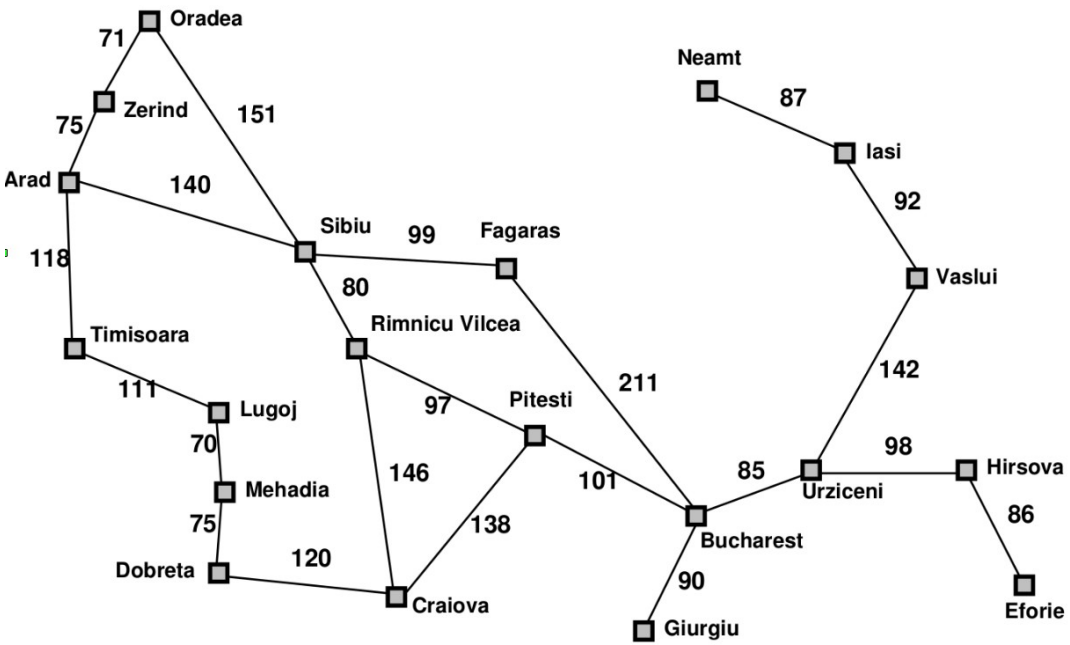

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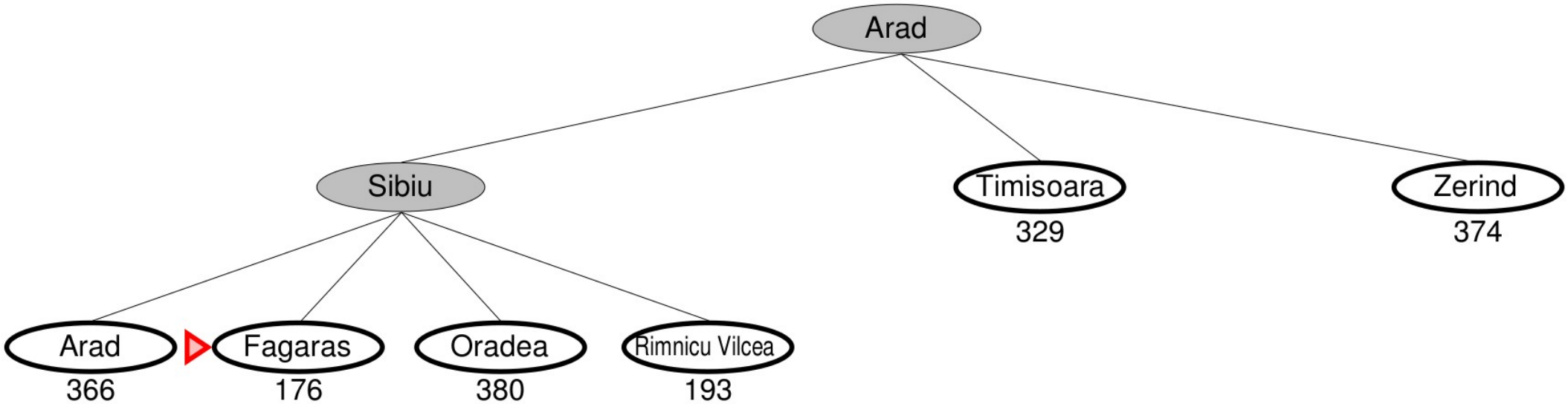

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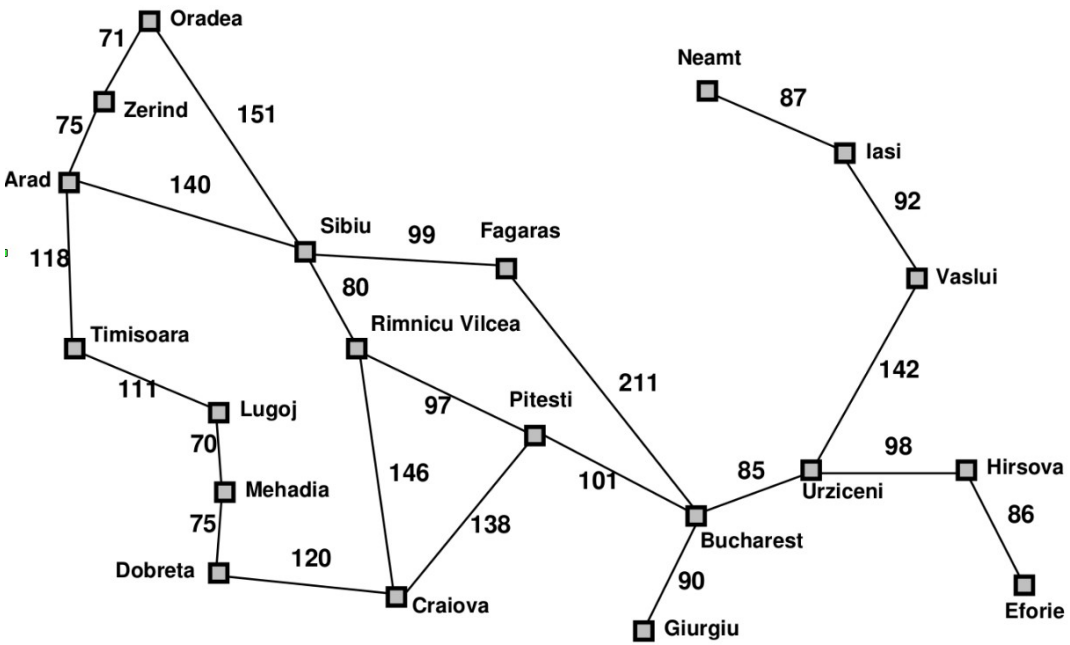

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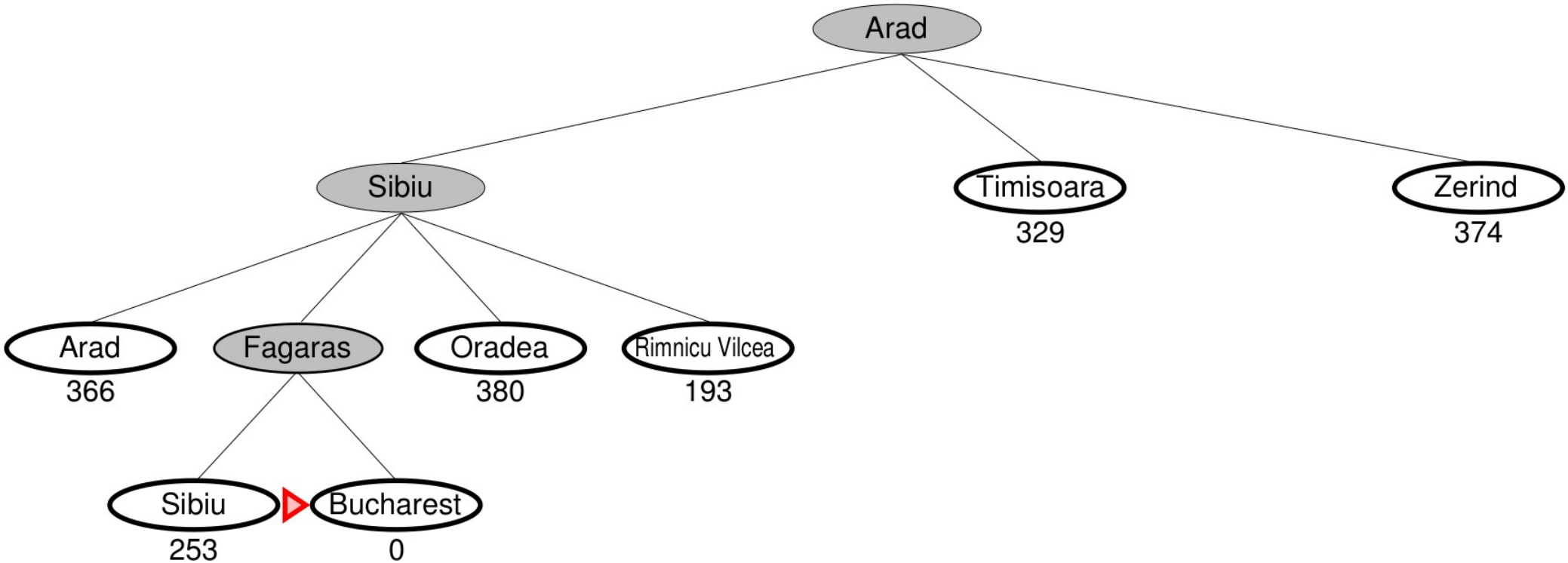
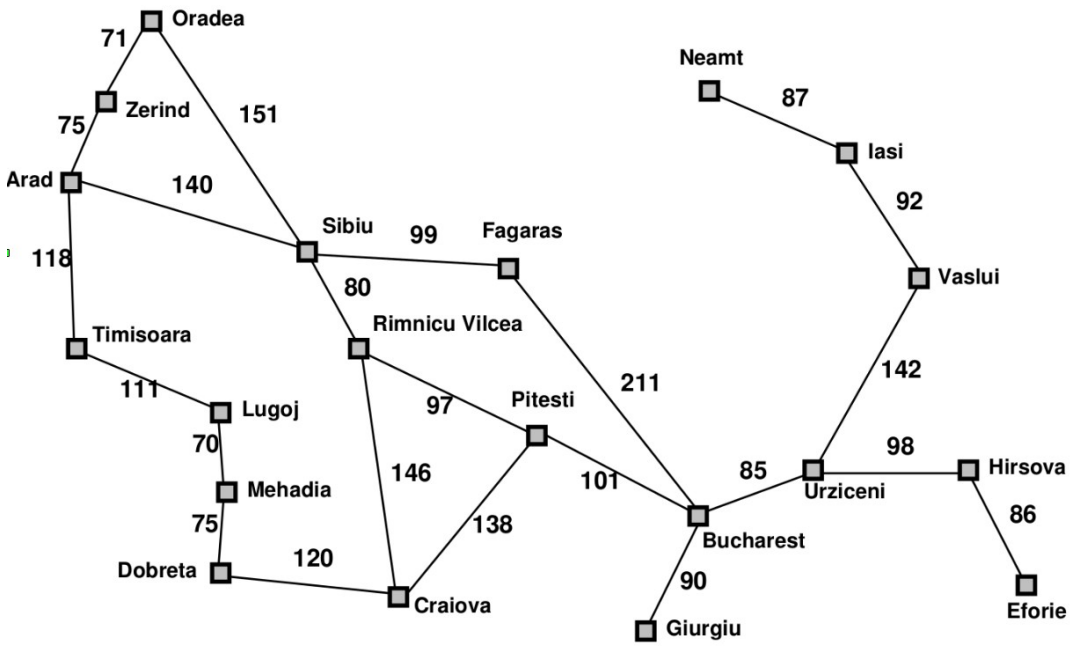


Recursive best-first search (RBFS)

- (RBFS) is **a** simple recursive algorithm that attempts to mimic the operation of standard best-first search, but using only linear space.
- Its structure is similar to that of a recursive depth-first search, but rather than continuing indefinitely down the current path it keeps track of the f-value of the best alternative path available from any ancestor of the current node.
- If the current node exceeds this limit, the recursion unwinds back to the alternative path.
- As the recursion unwinds, RBFS replaces the *f* -value of each node along the path with the best *f* -value of its children.

Recursive best-first search (RBFS)



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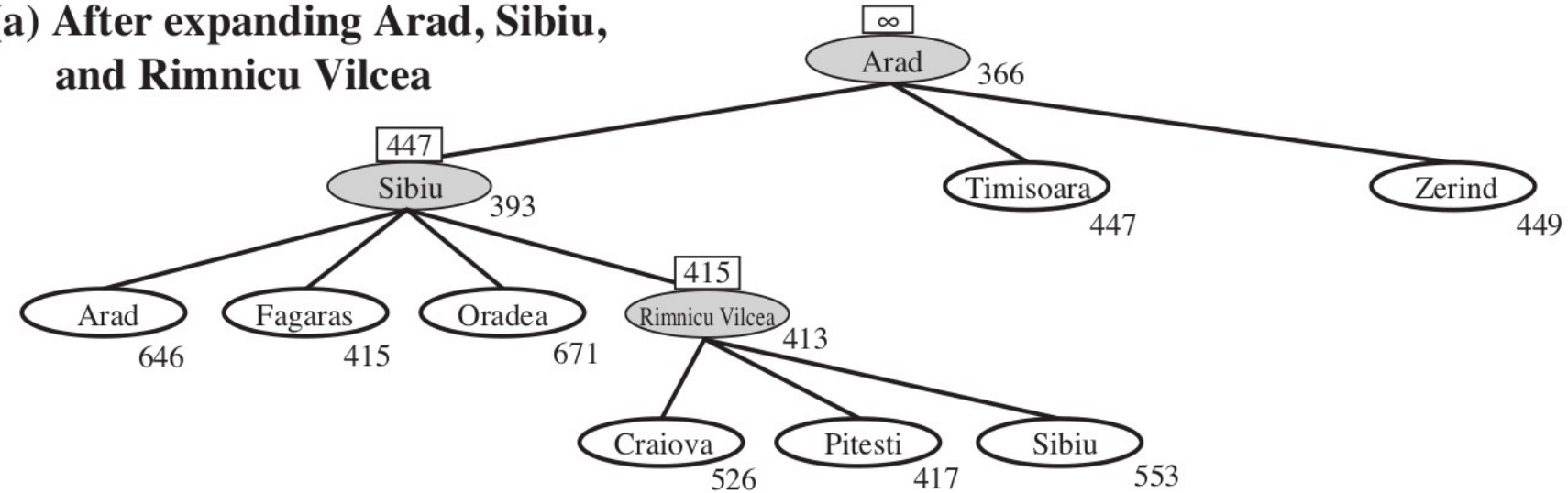
(a) After expanding Arad, Sibiu, and Rimnicu Vilcea



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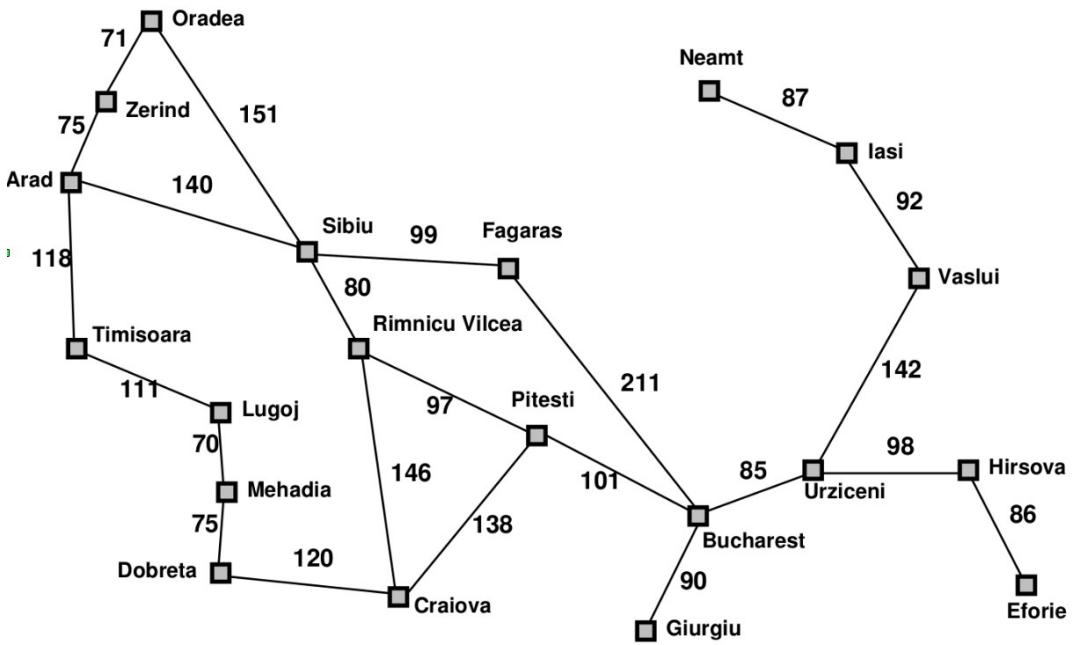

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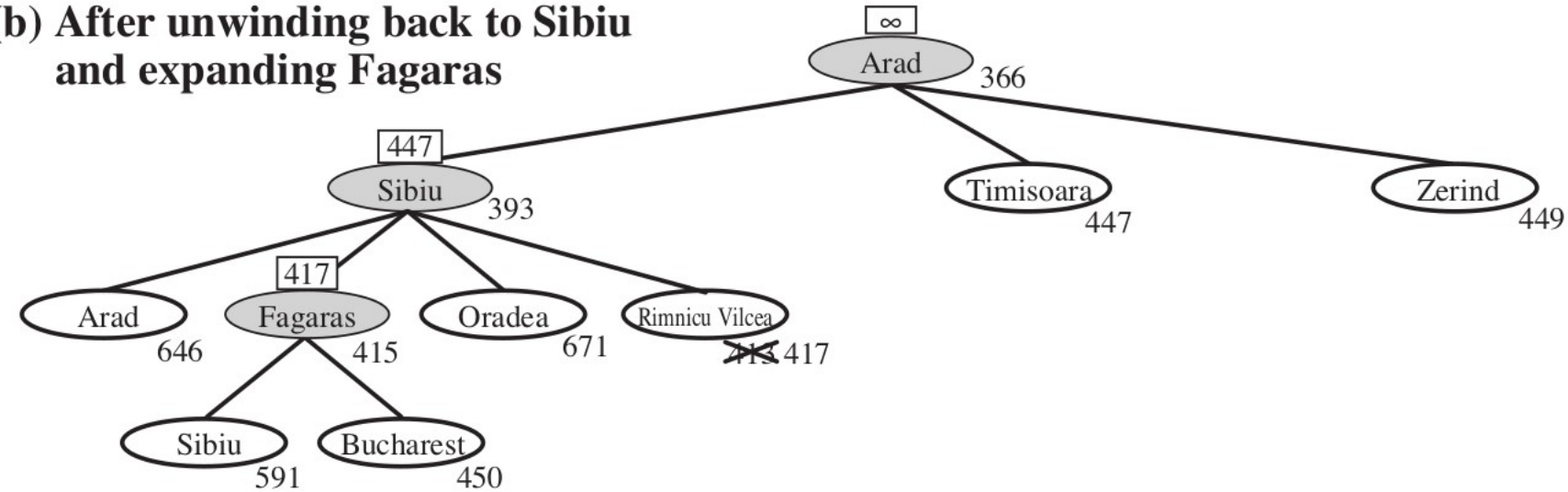
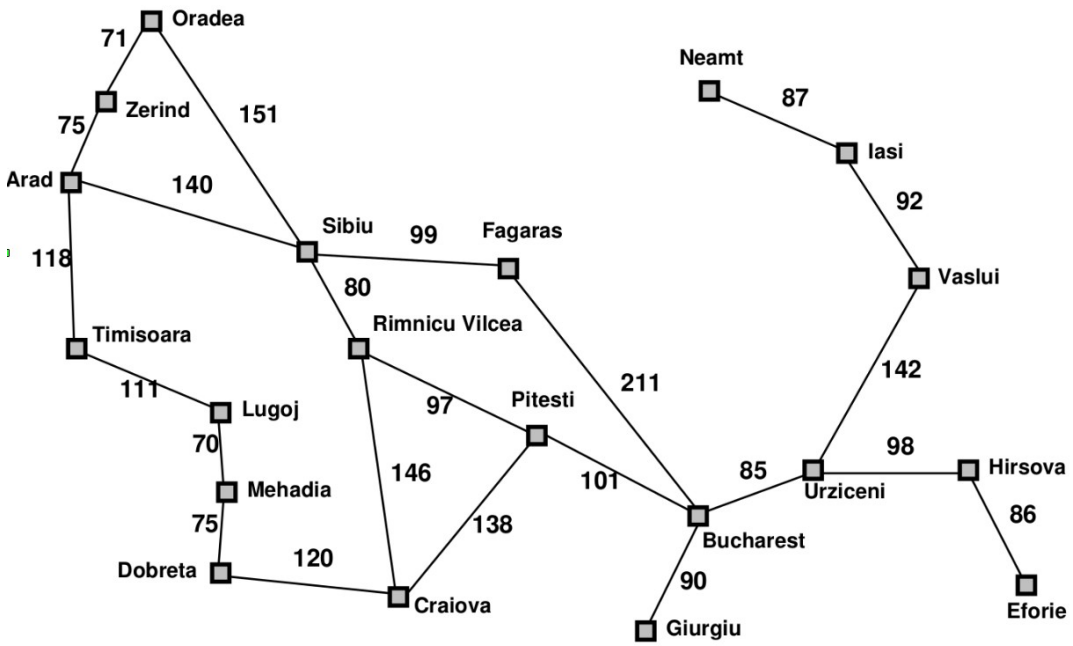
(b) After unwinding back to Sibiu and expanding Fagaras



Recursive best-first search (RBFS)



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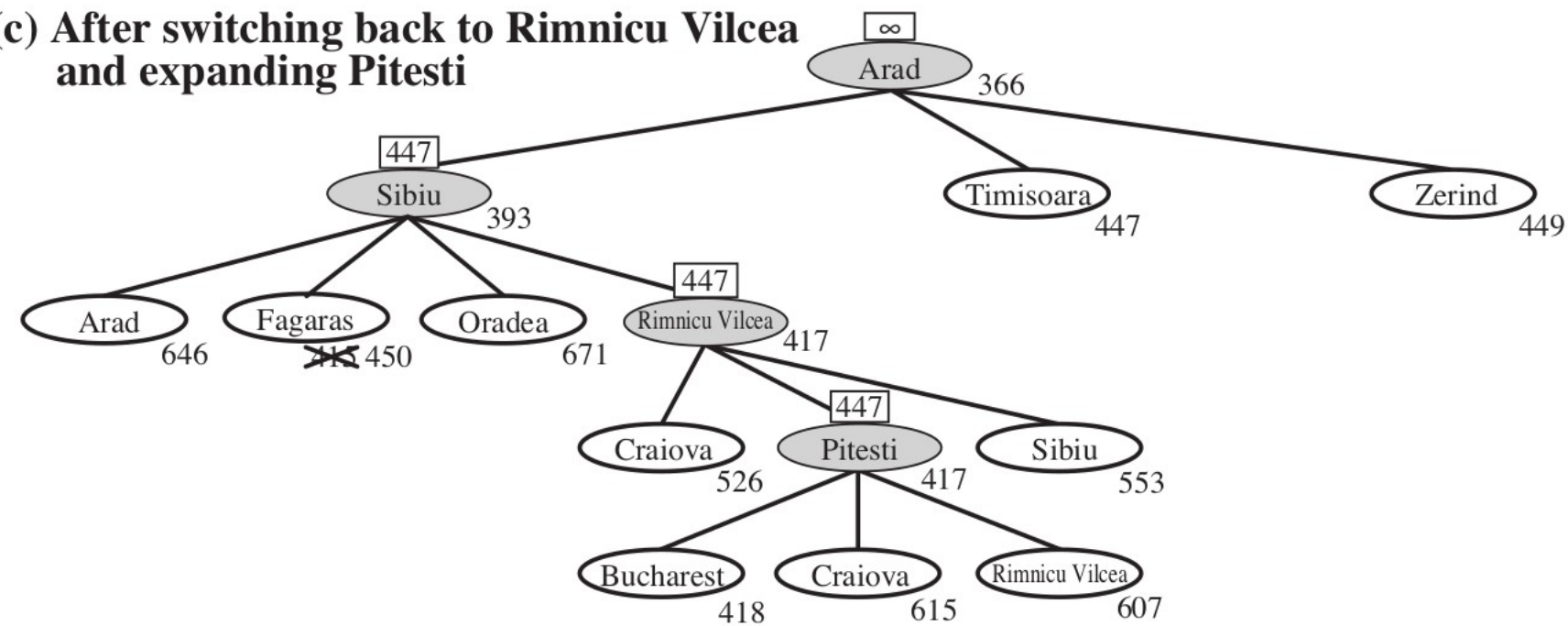
(c) After switching back to Rimnicu Vilcea and expanding Pitesti

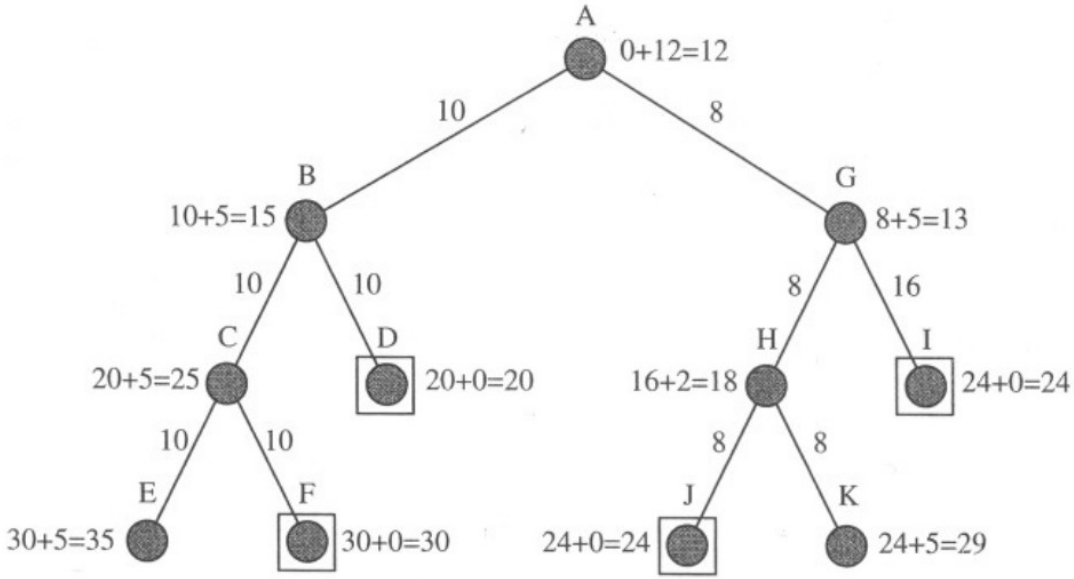


Recursive best-first search (RBFS)

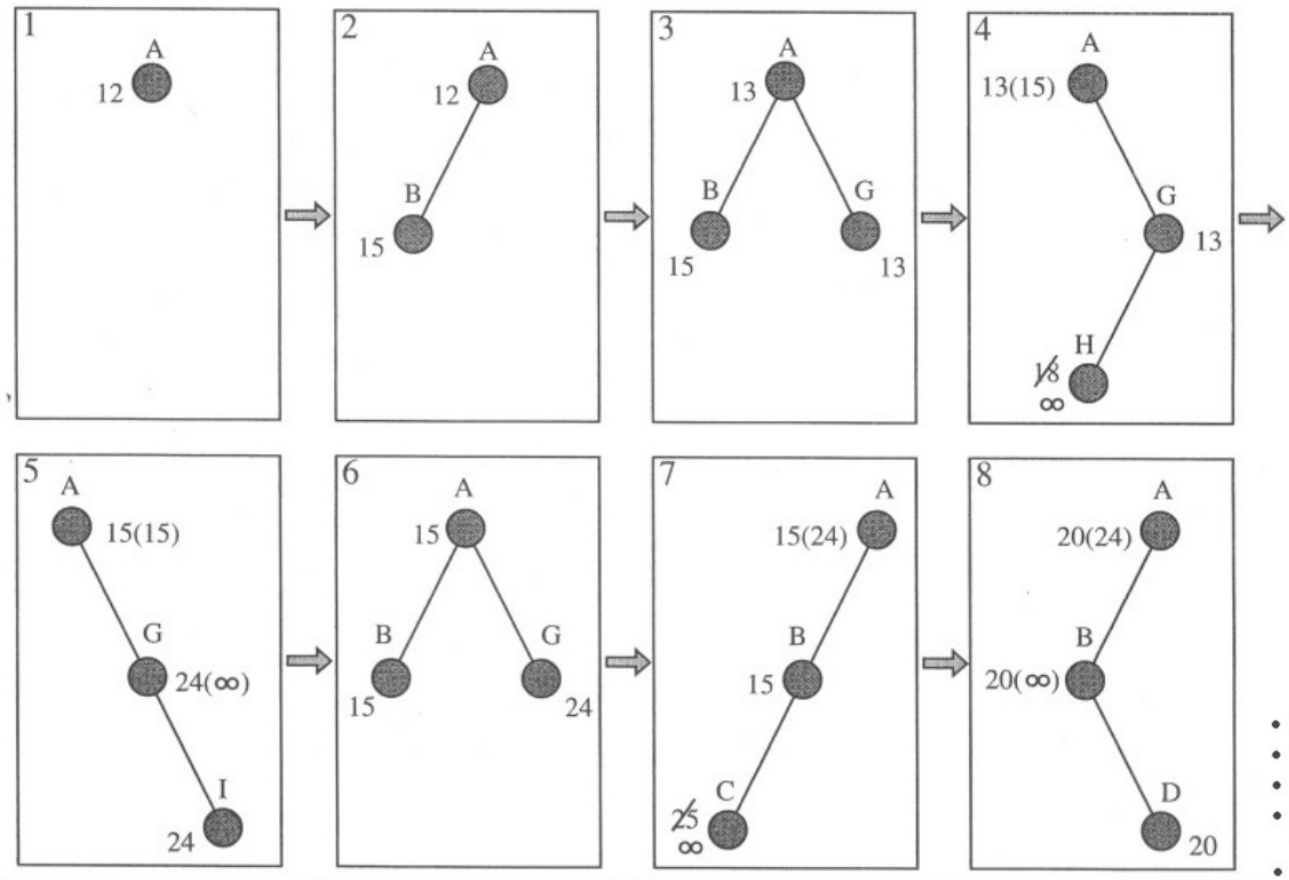
- RBFS is somewhat more efficient than IDA*, but still suffers from excessive node re-generation.
  - RBFS follows the path via Rimnicu Vilcea, then “changes its mind” and tries Fagaras, and then changes its mind back again.
  - These mind changes occur because every time the current best path is extended, its f -value is likely to increase—h is usually less optimistic for nodes closer to the goal.
  - When this happens, the second-best path might become the best path, so the search has to backtrack to follow it.
- Like A* tree search, **RBFS is an optimal algorithm if the heuristic function h(n) is admissible.**
- RBFS's **space complexity is linear in the depth of the deepest optimal solution**, but its time complexity is rather difficult to characterize: it depends both on the accuracy of the heuristic function and on how often the best path changes as nodes are expanded.

Drawbacks of RBFS & IDA*

- IDA* and RBFS suffer from using too little memory.
- Between iterations, IDA* retains only a single number: the current f -cost limit. RBFS retains more information in memory, but it uses only linear space: even if more memory were available, RBFS has no way to make use of it.
- Because they forget most of what they have done, both algorithms may end up reexpanding the same states many times over.
- Two algorithms that do this are MA* (memory-bounded A*) and SMA* (simplified MA*).



1. At each stage, one successor is added to the deepest lowest-f-cost node that has some successors not currently in the tree. The left child B is added to the root A.
2. Now $f(A)$ is still 12, so we add the right child G (f = 13). Now that we have seen all the children of A, we can update its f-cost to the minimum of its children, that is, 13. The memory is now full.
3. G is now designated for expansion, but we must first drop a node to make room. We drop the shallowest highest-f-cost leaf, that is, B. When we have done this, we note that A's best forgotten descendant has f = 15, as shown in parentheses. We then add H, with $f(H)$ = 18. Unfortunately, H is not a goal node, but the path to H uses up all the available memory. Hence, there is no way to find a solution through H, so we set $f(H) = \infty$.
4. G is expanded again. We drop H, and add I, with $f(I)$ = 24. Now we have seen both successors of G, with values of ∞ and 24, so $f(G)$ becomes 24. $f(A)$ becomes 15, the minimum of 15 (forgotten successor value) and 24. Notice that I is a goal node, but it might not be the best solution because A's f-cost is only 15.
5. A is once again the most promising node, so B is generated for the second time. We have found that the path through G was not so great after all.
6. C, the first successor of B, is a nongoal node at the maximum depth, so $f(C) = \infty$.
7. To look at the second successor, D, we first drop C. Then $f(D)$ = 20, and this value is inherited by B and A.
8. Now the deepest, lowest-f-cost node is D. D is therefore selected, and because it is a goal node, the search terminates.



SMA* search

- It will utilize whatever memory is made available to it.
- It avoids repeated states as far as its memory allows.
- It is complete if the available memory is sufficient to store the *shallowest* solution path.
- It is optimal if enough memory is available to store the shallowest optimal solution path. Otherwise, it returns the best solution that can be reached with the available memory.
- When enough memory is available for the entire search tree, the search is optimally efficient.

Summary

- **<u>Uninformed/Blind Search</u>**
  - ✔ DFS
  - ✔ BFS
  - ✔ Depth-limited search
  - ✔ IDS (or iterative deepening depth-first search)
  - ✔ Bidirectional
  - ✔ Uniform Cost
- **<u>Informed/Heuristic Search</u>**
  - ✔ Best-First Search
  - ✔ Greedy Best-First Search
  - ✔ A*
  - ✔ Memory Bounded Search
    - ✔ IDA*
    - ✔ RBFS
    - ✔ MA*
    - ✔ SMA*